

[illegible]

Title		
Size A2	Number	Revision
Date: 8/11/2026	Sheet of	
File: C:\Users\Tank and Door Module School	Drawn By:	

DOOR Sub-node - Schematic & Pin Map (all blocks except the RTD)

nRF54L15 (QFN52) | derived from Tank_and_Door_Schematic | SmartWagon Rev.1

Pin / net map

Function	Net	nRF54L15 pin	Dir	Firmware
I2C SDA (BMA400 + STS4X)	SDA	P1.04 / AIN0	bidir	i2c21
I2C SCL (BMA400 + STS4X)	SCL	P1.05 / AIN1	bidir	i2c21
Sensor-rail enable (TPS22917 ON)	SW_AXI	P1.06 / AIN2	OUT	srail.c
BMA400 interrupt (INT1)	BMA_INT	P0.04	IN	reserved
Debug UART TX	D_TX	P1.02 / NFC1	OUT	uart20
Debug UART RX	D_RX	P1.03 / NFC2	IN	uart20
Watchdog WAKE (TPL5010->MCU)	uC_Wakeup	P0.00	IN	watchdog.c
Watchdog DONE (MCU->TPL5010)	uC_Alive	P0.01	OUT	watchdog.c
Reset (TPL5010 RSTn, R17 pull-up)	NRESET	RESET	IN	-
LF crystal 32.768 kHz	XL1 / XL2	P1.00 / P1.01	-	LF XO
Door reed aggregate (RS1..RS4)	SW_1	P1.11 / AIN4	IN	door.c (pull-up)

Key components

Ref / Part	Bus / addr	Role
IC1 BMA400	I2C 0x14	Accel - tamper/impact + wagon-motion (door-open-while-moving)
U5 STS4X	I2C 0x4A*	Temperature - sub-node internal temp
RS1..RS4 reed	SW_1 net	Door open/closed (wired in parallel to GND)
U3 TPS22917	SW_AXI	Load switch - gates the VCC_AXI sensor rail
U4 TPL5010	R21=169k	External nano-power watchdog
BATT1 ER34615	-	3.6 V primary Li-SOCl2 cell

BLE payload mapping (struct sw_adv_pt)

Field	Meaning
value	STS4X internal temperature x10 (degC)
value2	peak acceleration x100 (g) -> tamper magnitude
flags	DOOR_OPEN IMPACT ALARM LOWBATT

Assumptions to confirm on the board

- ALARM = (door open AND wagon moving) OR impact. Open-while-stopped is reported as DOOR_OPEN only;
- the gateway classifies authorised/unauthorised using its geofence + motion context.
- STS4X I2C address 0x4A assumed - confirm the exact STS4x variant (0x44/0x45/0x4A/0x4B).
- Reed switches RS1..RS4 are wired in PARALLEL on one net (SW_1): firmware sees only the aggregate (line goes 'open' only when ALL reeds are open). Split onto separate GPIOs for per-door sensing.
- BMA_INT (P0.04) and SW_1 (P1.11) pin numbers taken from the U1B/U1C net labels - verify against the PCB.
- TPL5010 interval set by R21=169k - service DONE well within it (firmware feeds every report cycle).